



CODECHEF

Welcome to the CodeChef Challenge—your monthly dose of coding puzzles from India's biggest online programming contest, now in print!

Well, it's time to again compete with like-minded folks who are as passionate about programming as you are. Each month, we will bring to you a CodeChef challenge that will test your tech skills, and the top entries will be awarded by *LINUX For You* and Directi.

Last month's puzzle

How many squares of different sizes, and with sides parallel to the sides of the chess board, can we get from a 8x8 chess board?

The solution

An 8x8 chess board has 204 squares of different sizes with sides parallel to the sides of the chess board.

How we got the answer?

Answer is equal to $F(1) + F(2) + F(3) + \dots + F(8)$, where $F(n)$ is the number of squares with side 'n' units on the chess board.

Consider the top left corner of each square. The position of the top left corner along with 'n' can completely define a square. If the side of the square to be placed is 'n' units, then we can place the top left corner in $(8 - n + 1)$ ways on any row. So, for 'n' = 8, we can place the top left corner in $(8 - 8 + 1)$, i.e. 1 way on any row.

The same is true for the columns.

So, for a 8 x 8 square, we can use only the first row and column, for 7 x 7 squares, we cannot place the top left corner on any row or column number beyond 2.

Thus, the total number of squares of side 'n' units is $(8 - n + 1) \times (8 - n + 1)$. This is also the value for $F(n)$. Using this formula, the values of $F(n)$ for different values of 'n' are:

$$F(1) = (8 - 1 + 1) \times (8 - 1 + 1) = 64$$

$$F(2) = (8 - 2 + 1) \times (8 - 2 + 1) = 49$$

$$F(3) = (8 - 3 + 1) \times (8 - 3 + 1) = 36$$

$$F(4) = (8 - 4 + 1) \times (8 - 4 + 1) = 25$$

$$F(5) = (8 - 5 + 1) \times (8 - 5 + 1) = 16$$

$$F(6) = (8 - 6 + 1) \times (8 - 6 + 1) = 9$$

$$F(7) = (8 - 7 + 1) \times (8 - 7 + 1) = 4$$

$$F(8) = (8 - 8 + 1) \times (8 - 8 + 1) = 1$$

Add up all the results and you get 204

And the winners are...

- Wallace Jacob
- Anuvrat Parashar
- Arjun Pakrashi

Puzzle of the month

How many numbers from 1 to 10000 (inclusive) are divisible by 3 or 5 or 7?

Want to win?

Answer the question of the month for the CodeChef Challenge and mail it to codechef@efyindia.com. Three lucky winners will win Rs 1,000 each in cash! 

About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. CodeChef has been created by Directi, as a way to continuously challenge and engage the developer community. The goal is to provide a platform for practice, competition and improvement as well as enable developers to benchmark their skills against their peers. Log on to www.codechef.com for the December Challenge that takes place from December 1-11, 2009, to win cash prizes up to Rs 55,000!